PHASE CHANGE MATERIAL BASED PASSIVE COOLING SYSTEM FOR BATTERY THERMAL MANAGEMENT

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Abstract

This project investigates the design of an effective Battery Thermal Management System (BTMS) for electric vehicles by analyzing the thermal and electrochemical performance of Li-ion batteries. The widespread adoption of Li-ion batteries in EVs necessitates robust cooling solutions to prevent thermal runaway, a critical safety issue arising from excessive heat generation under extreme operating conditions. This study proposes an efficient and economical passive cooling system utilizing Phase Change Material (PCM). Three-dimensional simulations of a single battery cell (1P1S) and a six-cell battery pack (1P6S) were conducted using ANSYS Fluent. The simulations, which incorporated the multi-scale multi-dimensional NTGK model, evaluated the system's performance with and without PCM encapsulation. The NTGK framework was coupled with a solidification and melting model to accurately capture the thermal behaviour of the PCM. By analyzing the system's response to various discharge rates and ambient temperatures, this research demonstrates the effectiveness of the proposed PCM-based BTMS in mitigating temperature rise and ensuring safer battery operation.

1.Introduction

The global shift towards sustainable energy has accelerated the adoption of Electric Vehicles (EVs) and Hybrid Electric Vehicles (HEVs). At the heart of these vehicles are Li-ion batteries, favoured for their high energy density and long cycle life. However, the performance and safety of these batteries are highly sensitive to temperature. During operation, especially under high discharge rates or in high ambient temperatures, Li-ion batteries generate significant heat due to electrochemical reactions and internal resistance. If this heat is not effectively dissipated, it can lead to a dangerous condition known as thermal runaway, where a chain reaction of exothermic events can cause fire and explosion [5].

To ensure safe and efficient operation, a Battery Thermal Management System (BTMS) is essential. A well-designed BTMS must maintain the battery temperature within an optimal range (typically 20-40°C) and ensure temperature uniformity across the pack (ideally with a variation of less than 5°C) [9]. Common cooling methods include air cooling, liquid cooling, and passive systems using Phase Change Materials (PCMs). While active systems like liquid cooling are highly effective, they add complexity, weight, and parasitic power loss to the vehicle.

PCM-based systems offer a promising alternative. They function passively by absorbing latent heat as the material melts, thereby regulating the battery temperature without requiring external power. This makes them an economical, lightweight, and reliable solution. This project focuses on developing and analyzing a PCM-based BTMS for a Li-ion battery pack through detailed numerical simulation.

2.Research Focus

This project focuses on simulating PCM-based cooling system for Li-ion battery.By using ANSYS Fluent software ,simulations were conducted on various battery models (prismatic battery 1p1s and pack 1p6s with and without pcm) working within temperature range of 20°C-50°C under different C-rates.From the results of simulations,we have

analyzed temperature distributions and voltage curves.This valuable data can be used to optimize battery performance and safety.

3. Literature Overview

A comprehensive review of existing literature was conducted to inform the methodology of this study. This body of work provides a strong foundation for the numerical modeling approach adopted in the present study. Key findings from relevant research are summarized below:

1.**Hussain et al. [1]** have shown that while increasing coolant flow rates in liquid cooling systems can reduce peak temperatures and temperature non-uniformity, it comes at the cost of higher energy consumption.

2.**Choudhari et al. [2]** have demonstrated that PCM cooling is effective for moderate discharge rates (1C/2C), but its performance diminishes at higher rates due to the finite heat storage capacity of the material.

3.**Hwang et al. [3]** explored methods to enhance the thermal conductivity of PCMs, such as using carbon-based fillers or heat sinks, to address the limitations of passive systems.

4.**Vermeer et al. [4]** highlighted the link between anode potential and the growth of the Solid Electrolyte Interphase (SEI) layer, which is a key factor in battery aging.

5.**Gibbard et al. [5]** and **Yazami et al. [6]** have provided thermodynamic frameworks for understanding the energy and entropy changes that occur during electrochemical reactions.

6.**Gu et al. [7]** showcased the development of multi-dimensional thermal-electrochemical models that have enabled the accurate prediction of internal temperature distributions.

7.**Basu et al. [8]** identified contact resistance as a dominant factor in the thermal performance of a battery pack.

8.**Bahiraei et al. [9]** used electrochemical-thermal modeling to show how the geometry of cooling plates significantly affects temperature gradients within a battery module.

9.**Hussain et al. [1]** also found that for a PCM encapsulated battery pack, the strength of convective effects in the liquid phase of the PCM is negligibly small, with an average temperature deviation of only 0.25%. This finding suggests that PCM can be accurately modeled by considering only conduction, simplifying the simulation.

10. **Kim et al. [10]** conducted experimental studies on the thermal behavior of LMO-cathode batteries, providing valuable data for validating numerical models.

11. **Ceyda & Alkaya [11]** performed numerical investigations into the thermal behavior of Li-ion batteries under different load conditions, offering another key source for model validation.

12. Standard numerical practices, such as those described by **Ferziger & Perić [12]**, are essential for ensuring the stability and convergence of complex fluid dynamics simulations.

13. Commercial CFD software documentation, such as the **ANSYS Fluent Theory Guide [13]**, provides the theoretical basis for the models and algorithms used in the simulation.

4 BTMS:PCM-Based Cooling System

4.1 Objective

1.To analyze battery thermal performance by numerical methods (NTGK model) considering only natural convection at different C-rates of single prismatic cell without PCM.

2.To perform simulations on single battery and battery pack (1p6s) considering PCM .

3.To plot different curves such Maximum Temperature vs time, $\triangle T$ vs Time, liquid fraction vs time and cell voltage vs discharging capacity and cell voltage vs time.

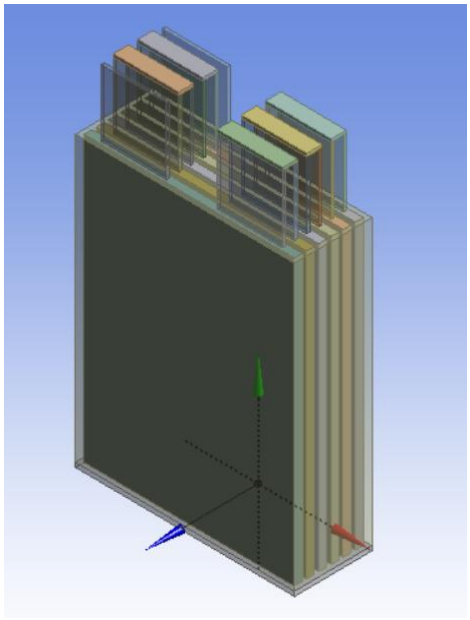
4.2 Numerical Methods

The simulations were performed using the commercial solver ANSYS Fluent. The thermal and electrochemical behavior of the battery was modeled using the multi-scale multi-dimensional (MSMD) Newman, Tiedemann, Gu, and Kim (NTGK) model. The NTGK model is a semi-empirical framework that accurately predicts heat generation by relating volumetric current density to cell voltage and depth of discharge [7].

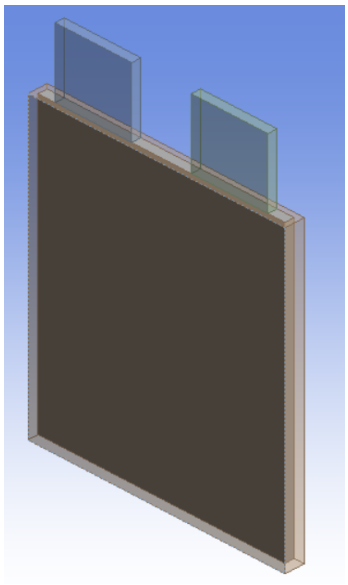
To simulate the behavior of the PCM, the NTGK model was coupled with ANSYS Fluent's built-in solidification and melting model. This model utilizes an enthalpy-porosity formulation to track the phase transition of the PCM, accurately accounting for the latent heat absorbed during melting.

4.3 Model Geometry

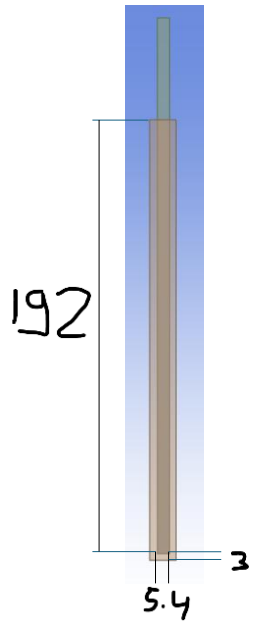
The study involved simulations of both a single Li-ion pouch cell (1P1S configuration) and a battery pack composed of six identical cells arranged in series (1P6S configuration). The geometry and material properties of the battery, which features a Lithium Manganese Oxide (LMO) cathode, were based on the work of Ceyda et al. [11]. For the thermal management system, the battery pack was encapsulated in a layer of Phase Change Material (n-octadecane), as depicted in Fig 1.The battery has dimensions of 145×5.4×192 in mm with pcm of thickness 3mm.Dimensions and Physical parameters of battery are given in Table 1.The properties of battery,tab material and PCM is given in Table 2.



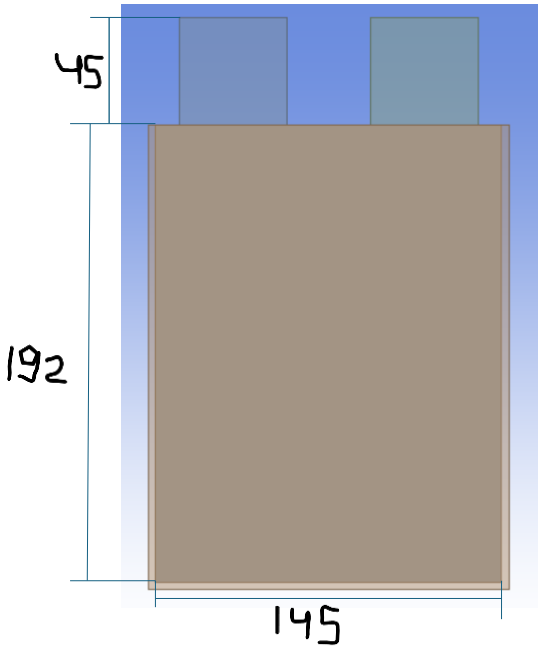
(a)



(b)



(C)



(d)

Fig 1. (a) Battery pack with pcm. (b) Isometric view of single battery with pcm. (C) Side view. (d) Front view of battery.All dimensions are in mm

Table 1.

Physical parameters of a battery pack.

Parameter name	For 1P1S battery
Cathode material	LMO
Anode material	Graphite
Electrolyte	Polymer-based
Nominal voltage	3.7 V
Maximum cut-off potential	4.3 V
Minimum cut-off potential	3 V
Nominal capacity	14.6 Ah

Table 2.

Properties of battery, PCM and tab materials.

Parameters	Battery(Li-ion) [11]	Terminals/tables [11] (Cu)	PCM (n-octadecane)	
			Solid	Liquid
Density ρ (kg/m3)	2092	8978	814	724
Specific heat cp (J/kg-K)	678	381	2150	2180
Thermal conductivity k (W/m-K)	18.2	387.6	0.358	0.152
Melting heat (kJ/kg)	__	__	225	
Solidus temperature TS (K)	__	__	301.15	
Liquidus temperature TL (K)	__	__	303.15	

4.4 Mesh quality and time- step

A highly structured mesh is generated for all the designs involving the single battery and battery pack with and without PCM encapsulation. For non-encapsulation designs, the quality of the mesh averaged at 0.96, and the orthogonality of 0.97 are the indicators of an excellent mesh. In the case of encapsulated designs, the mesh quality remained at an average of 0.91, and the orthogonality of 0.93 is considered very good [13], as shown in **Fig. 2**.Three different meshes has been considered by generating a mesh of 20k ,192k and 500k nodes for 1P1S with pcm encapsulation and 11k,89k and 213k without pcm .From the results mesh of node 192k with pcm and 89k without pcm encapsulation was taken for further simulations because there was not much difference between meshes

2nd and 3rd in increasing order of nodes for both cases with pcm and without pcm.similarly ,time step of 2.5 was selected for further calculations to reduce calculation time .When simultions were performed on 1p1s battery at 5c rate and ambient tempertature of 300k,time step of 0.5,1,2.5 and 10s was considered but there was not much difference in results of time step 2.5s and 0.5s .

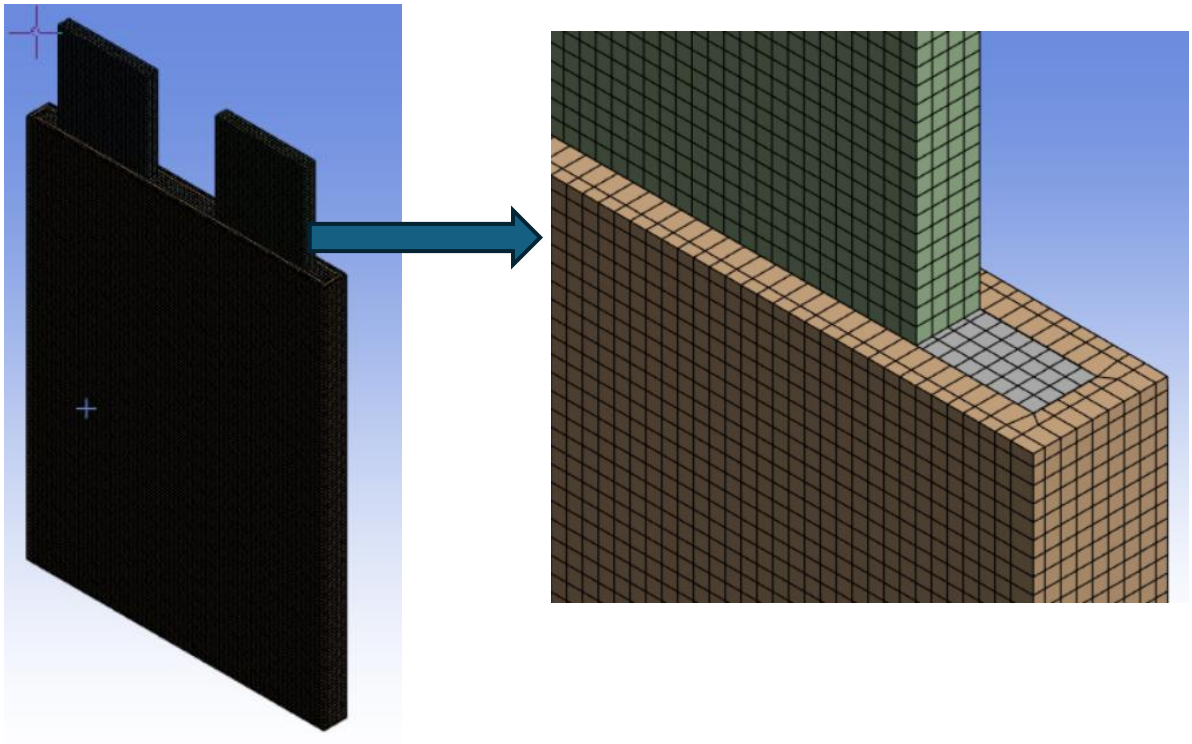


Fig2.meshing of 1p1s battery

4.5 Model validation

To ensure the reliability of the numerical setup, the battery model was first validated against existing data. The simulated discharge curves were compared with the experimental results from Kim et al. [10] and the numerical results from Ceyda et al. [11]. As shown in Figure 3, the results from the present study show excellent agreement with the reference data, with a mean deviation of only 1.14% from the experimental work and 0.42% from the numerical work. This successful validation confirms the accuracy of the modeling approach.

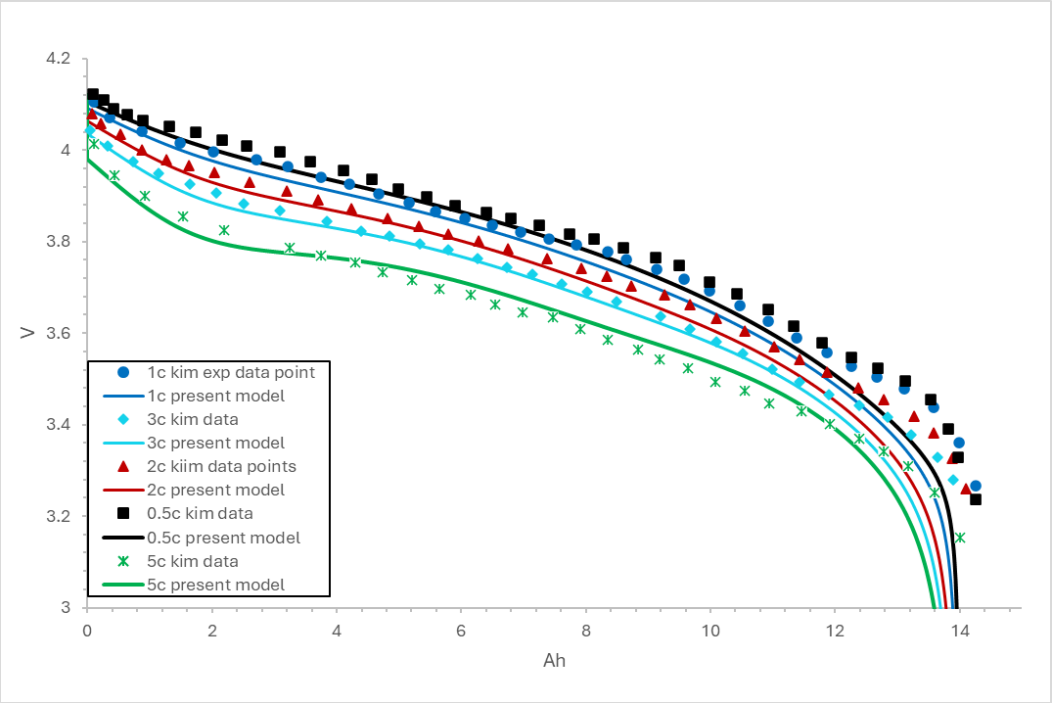


Fig 3(a).model validation with experimental study of Kim et al.

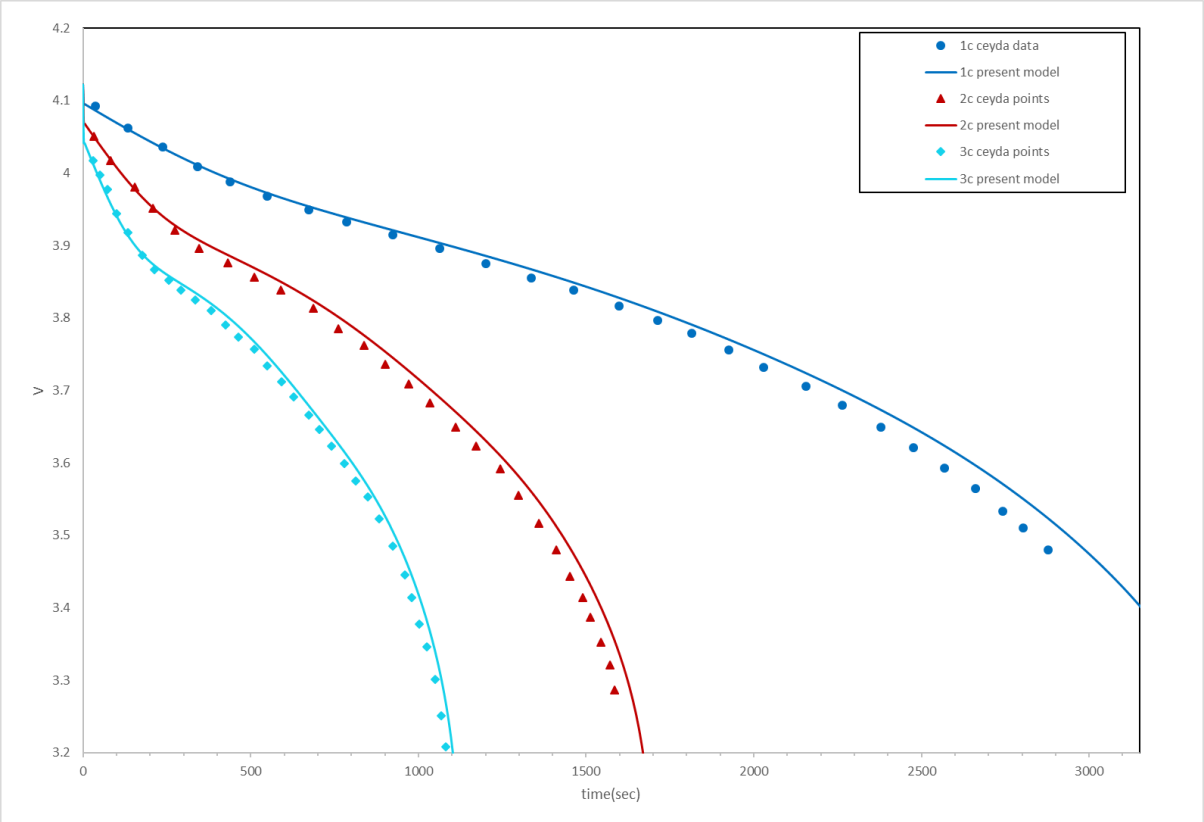


Fig 3(b). Model validation with numerical study of Ceyda et al.

5.Results and Discussion

The performance of the PCM-based BTMS was evaluated under various discharge rates. The analysis focused on the maximum temperature reached within the battery pack and the temperature uniformity.

5.1 For 1P1S battery with and without PCM at different ambient temperature

Figure 4 illustrates the effect of ambient temperature on the maximum temperature of a 1P1S battery with and without PCM, along with the variation in temperature difference (ΔT) and the average liquid fraction of the phase change material (PCM) at a 5C discharge rate. The plot clearly indicates that the PCM is effective only when phase change occurs at the given ambient temperature.

In this study, n-octadecane is used as the PCM, which has a melting temperature range of 301.15–303.15 K. When the ambient temperatures are 293 K and 300 K, the PCM remains in its solid state initially and undergoes a phase change during operation, thereby absorbing heat generated by the battery. However, at higher ambient temperatures such as 313 K and 323 K, n-octadecane is already in a fully liquid state, rendering it ineffective for thermal management as no phase change—and hence, no latent heat absorption—occurs.

During the phase transition, the temperature within the battery remains nearly constant, leading to a significant drop in the temperature difference across the battery. The liquid fraction of the PCM varies from 0 (completely solid) to 1 (completely liquid), indicating the extent of phase change. From Fig 4(a). it is cleared that PCM has lowered the max temperature of battery from 322.1K to 302.9K,326.76K to 303.4K,336.03Kto 328.1K and 343.5K to 336.2K at ambient temperature of 293K,300K,313K and 323K respectively.

As shown in **Figure 4(b),** at an ambient temperature of 293 K, the temperature difference initially increases because the PCM is solid and begins absorbing heat only after reaching its melting point—approximately four minutes into operation. Once melting starts, the battery temperature stabilizes, as reflected in the plot. A similar thermal behaviour is observed at an ambient temperature of 300 K.Fig.4(c) shows liquid fraction of PCM ,it is cleared from plot liquid fraction increases with increase in ambient temperature and it is 1 if ambient temperature is more than 303.15K for n-octadecane.

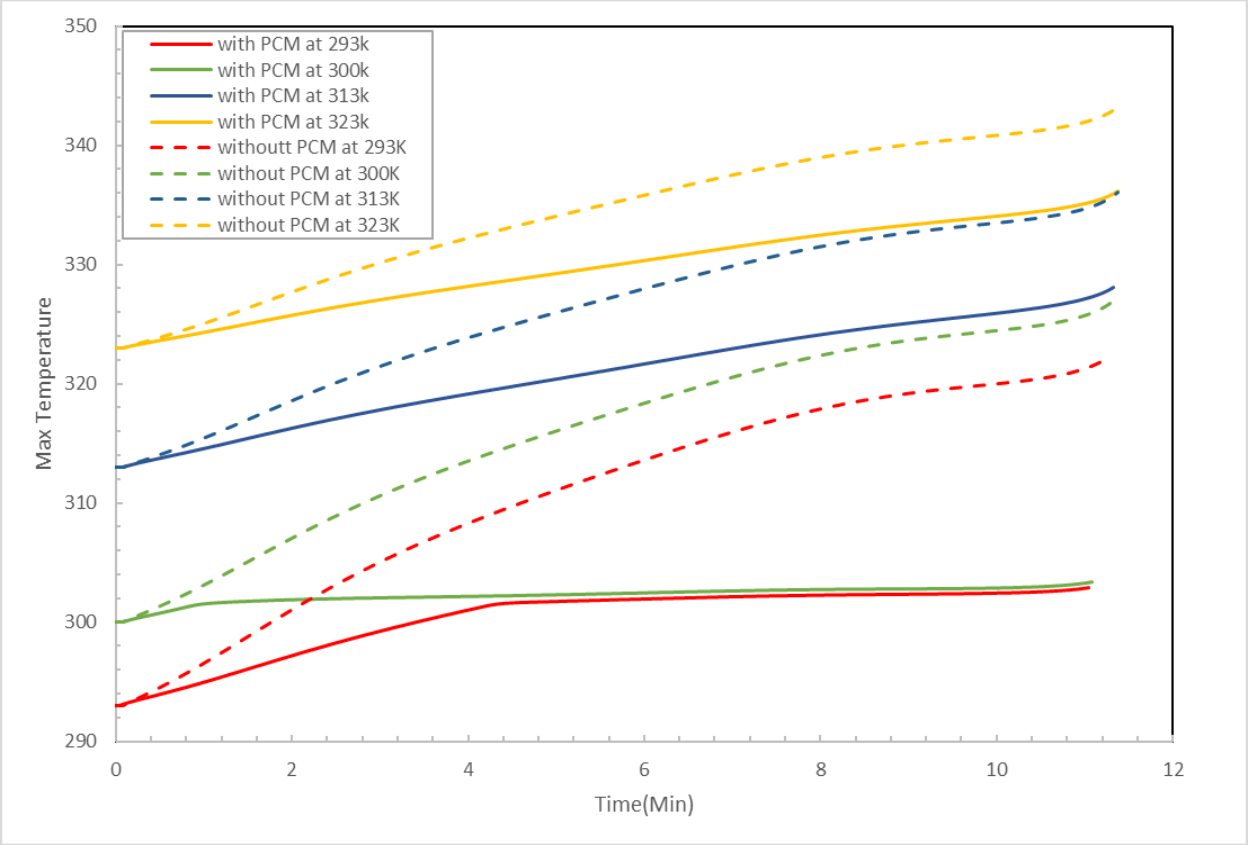
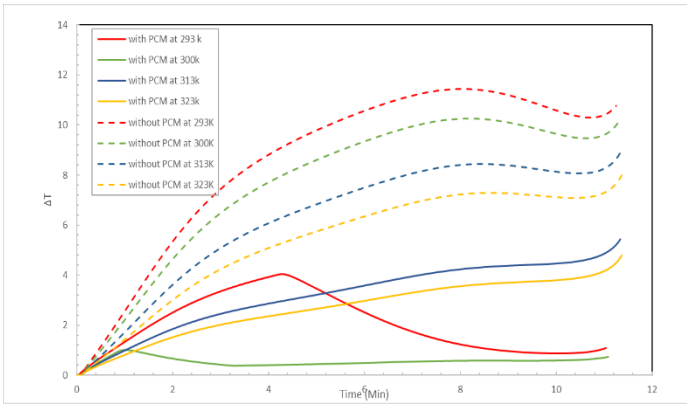
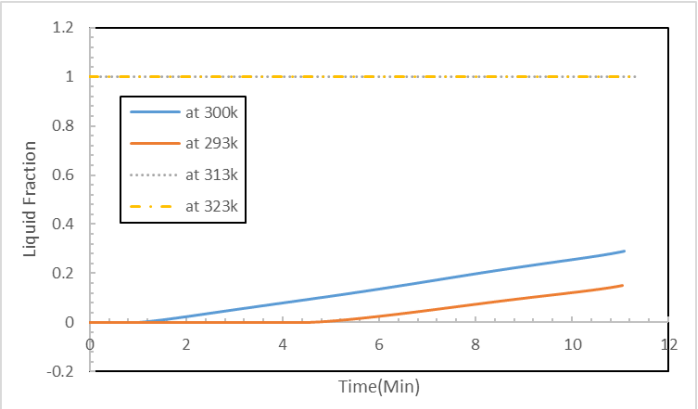


Fig.4.When 1P1s battery discharged at different ambient temperature at 5C rate with and without PCM(a) Max Temperature(K) vs Time.



(b) ΔT vs Time .



(c)Liquid fraction vs Time.

5.2 For 1P6S battery with and without PCM at different ambient temperature

5.2.1Effect of Discharge Rate on Temperature

Figure 5. illustrates the maximum temperature within the 1P6S battery pack over discharged capacity(Ah) for different discharge rates (1C, 2C, 3C, and 5C), comparing the performance with and without PCM encapsulation. Without the PCM, the battery temperature rises significantly, especially at higher C-rates. At a 5C discharge rate, the maximum temperature approaches 327.64 K. However, with the PCM encapsulation, the temperature rise is substantially mitigated. The PCM absorbs the excess heat generated during discharge, maintaining the maximum temperature at a much lower and safer level, around 303.97 K. This demonstrates the effectiveness of the PCM in preventing overheating**.**

5.2.2 Temperature Uniformity

Maintaining a uniform temperature distribution is crucial for preventing localized degradation and ensuring a long battery life. Figure6 shows the temperature difference (ΔT) across the battery pack over discharged capacity(Ah). At 5C rate, Without PCM, the temperature difference quickly exceeds the recommended 5 K limit, indicating significant thermal imbalance. In contrast, the pack with PCM maintains a temperature difference well below 5 K throughout the discharge cycle. This shows that the PCM not only reduces the peak temperature but also distributes the heat more evenly across the pack.

5.2.3 PCM Liquid Fraction

The effectiveness of the PCM is directly related to its phase transition. Figure 7 shows the liquid fraction of the PCM over DOC at different discharge rates. The liquid fraction indicates the proportion of the PCM that has melted. At a low discharge rate of 1C, the heat generated is minimal, and only a small fraction of the PCM melts. As the discharge rate increases to 3C and 5C, the higher heat generation causes the PCM to melt more rapidly and completely. This rapid phase change allows the PCM to absorb a large amount of latent heat, thereby stabilizing the battery's temperature even under high thermal

5.2.4 cell voltage

Figure 8 shows pack voltage vs discharged capacity ,from the plots it clear that voltage at same discharged capacity without PCM encapsulation is high at same C rate when compared with encapsulation of pack .The main reason for this is that when there is no encapsulation ,temperature of pack is high and therefore working voltage is high as

compared when it is encapsulated with PCM. Figure 9 shows total heat generated by battery pack with and without PCM encapsulation ,from the plot it cleared that heat generated is more at same C rate when Pack is encapsulated by PCM as compared to without PCM encapsulation.The reason behind this is that in PCM encapsulation effective heat transfer coffiecient is high due to PCM while when there is no PCM the only heat removed from battery pack is due to surrounding .

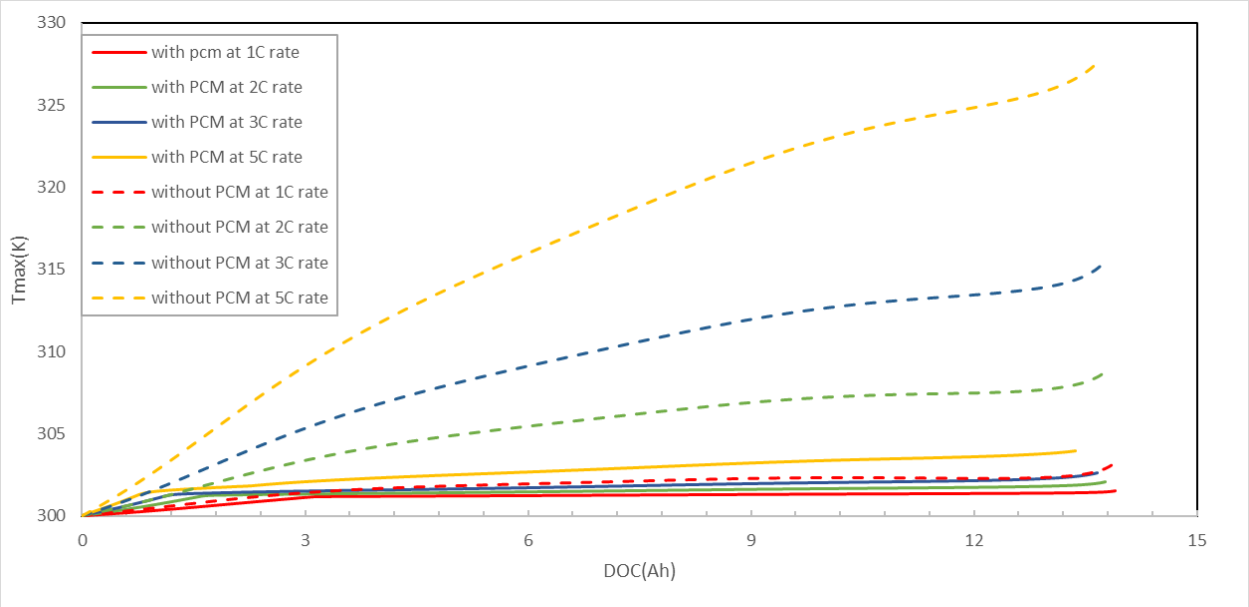


Fig.5.T max(K) vs discharged capacity with and without PCM encapsulation at different C rates at ambient Temperature of 300K.

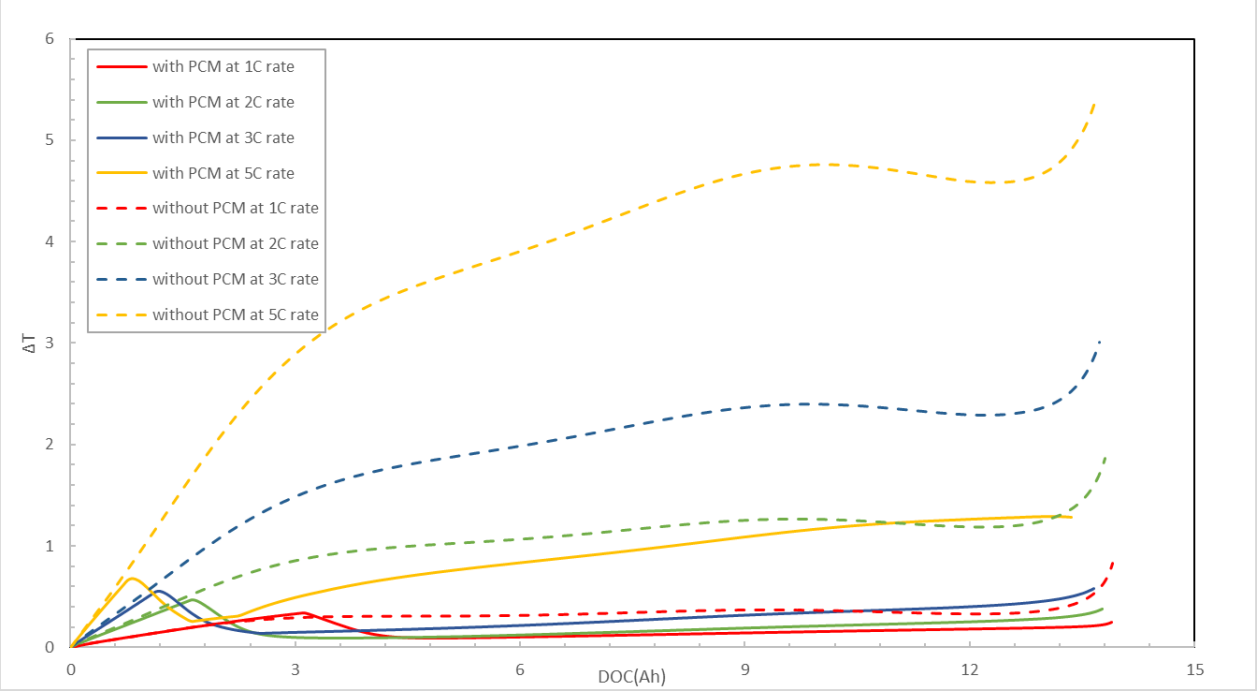


Fig.6. ΔT vs discharged capacity with and without PCM encapsulation at different C rates at ambient Temperature of 300K.

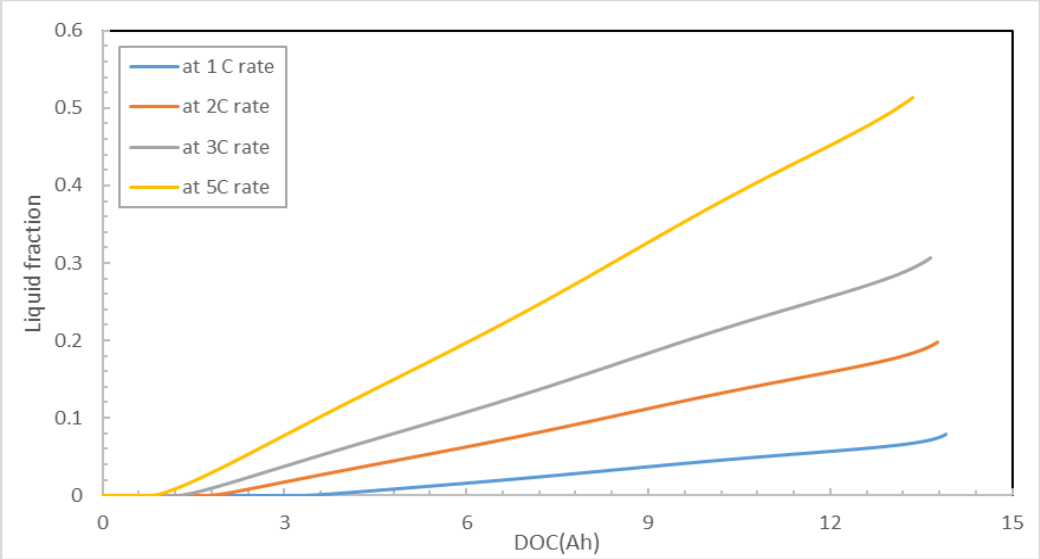


Fig.7. Liquid Fraction vs discharged capacity with and without PCM encapsulation at different C rates at ambient Temperature of 300K.

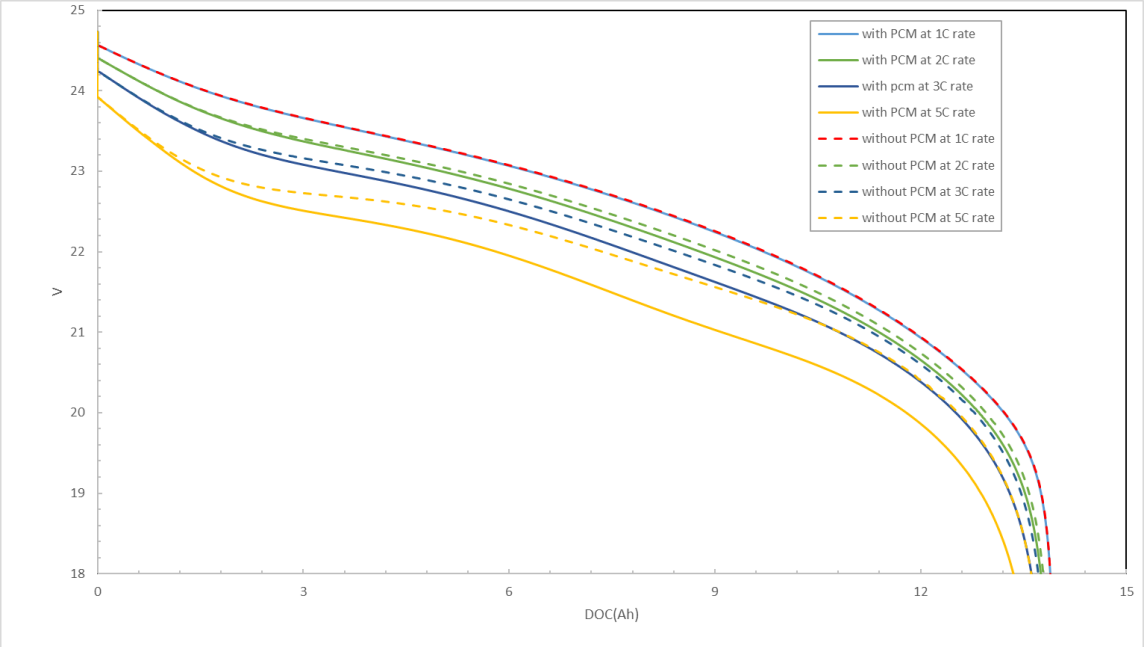


Fig.8.pack voltage vs discharged capacity with and without PCM encapsulation at different C rates at ambient Temperature of 300K.

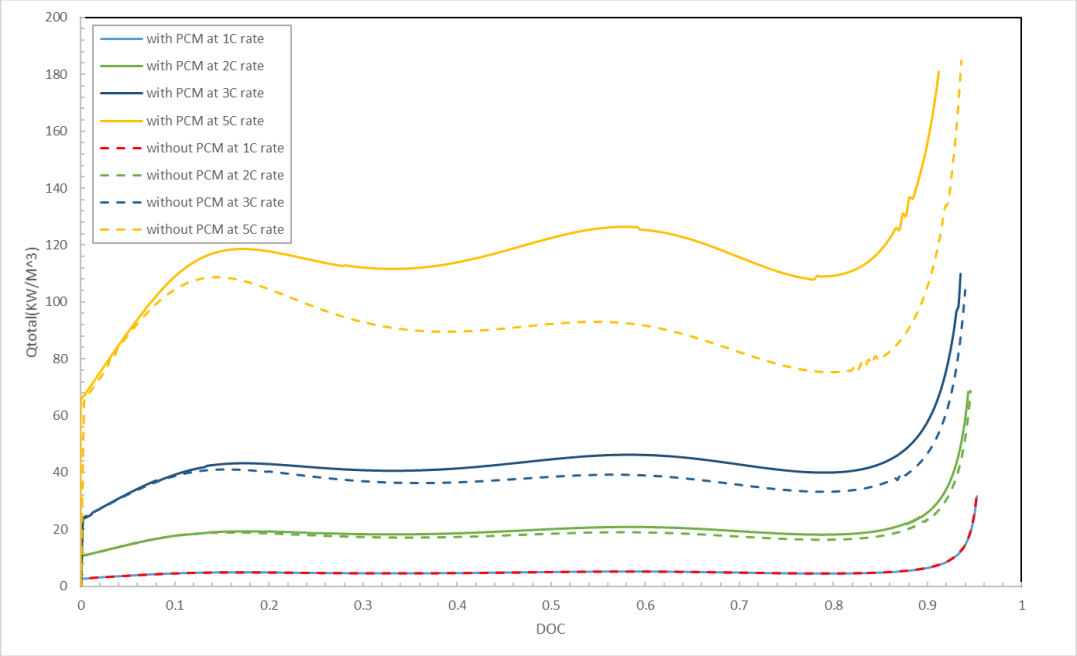


Fig.9.Total heat generated vs depth of charge with and without PCM encapsulation at different C rates at ambient Temperature of 300K.

6. Conclusion

This study successfully developed and validated a numerical model to analyze a PCM-based passive thermal management system for a Li-ion battery pack. The simulations demonstrate that the proposed system is highly effective at regulating battery temperature under various operating conditions. Key findings include:

- The PCM encapsulation significantly reduces the maximum temperature of the battery pack, preventing it from reaching critical levels, especially at high discharge rates.
- The system greatly improves temperature uniformity across the pack, keeping the maximum temperature difference below the recommended 5 K threshold.
- The analysis of the PCM's liquid fraction confirms that the material effectively absorbs thermal energy through its phase transition, providing robust passive cooling.

Overall, this research confirms that a well-designed PCM-based BTMS is an efficient, economical, and reliable solution for managing the thermal challenges in electric vehicle batteries, contributing to their safety and longevity.

7. References

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